CA/Browser Forum Tracker

Part 4 - Public Trust Programs

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About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. This tracker is a summarized pull of the public record, kept as a coherent, concise news update. It draws on the CA/Browser Forum's own ballots, mailing lists and published documents, and on the public sources of the browser root programs. An AI model reads those sources and writes the summary, and every statement links to the source it came from. Where two sources disagree, both are reported. AI makes mistakes, so check the linked source before relying on any statement. The tracker is updated each week that the public record shows a change.

The Forum writes the rules, and the root programs decide who is trusted. A root program can and does demand more than the Baseline Requirements, often a year or more ahead of them. This part covers that enforcement layer as it stood on 1 November 2026.

Section 1 covers the four root store policies. Section 2 follows the separate post-quantum trust path that Chrome and Apple are designing. Section 3 covers which authorities are entering and leaving the stores, and Section 4 covers the incident record. Section 5 covers the defining event of the autumn, the compromise of eight eMudhra intermediate keys and the removal of the root above them by three programs. Section 6 covers GoDaddy's revocation of more than 400,000 certificates, which is complete and now has its full incident report.

Two kinds of trust removal appear below and should not be confused. One follows a pattern of misconduct or a compromise and is protective. The other follows the age of a root and is routine housekeeping. Between September and 19 October no program took either kind of action.

On 20 October Mozilla took the first kind. Chrome and Microsoft followed, and by 1 November all three had removed the same root.

1. Root Store Policies

Four organisations run root programs that matter for the public web, and their policies are converging on dedicated hierarchies, bounded root lifetimes and stronger evidence of what an authority actually does.

Recent Developments

Mozilla Root Store Policy version 3.1 was published on 29 June 2026 and took effect on 1 July 2026 [1]. It requires policy and practice documents to be explicit, bounded, auditable and detailed enough to describe issuance [1]. It also requires root key pairs to have been generated within the previous five years [1].

Its largest change arrives later. For audit periods beginning on or after 1 July 2027, authorities with roots enabled for websites must obtain a Detailed Controls Report [1]. On 23 September Mozilla published a whitepaper on what those reports must contain [2]. On the same day it merged a template for mass revocation plans [3]. The template sets thresholds of 100 certificates or 1% of an authority's active web certificates [3].

The Chrome Root Program Policy is at version 1.8 [4]. It limits the number of roots per owner and requires policy documents to state adherence to the latest policy and to the CCADB Policy from 15 June 2026 [4]. Owners with more than two self-signed roots had to submit a consolidation plan by 15 June 2026, and the cap of two takes effect on 15 September 2027 [4].

Chrome is also removing client authentication from its hierarchies. New subordinate certificates disclosed from 15 June 2026 must carry server authentication only, and new end-entity certificates follow on 15 March 2027 [5]. That timetable comes from a secondary summary of the policy rather than the policy text itself [5].

Microsoft's requirements now cap new roots at ten years of validity [6]. Roots submitted from 1 July 2026 must be dedicated to one purpose, with separate roots for server authentication, email and code signing [6]. Apple's policy version 2.0 took effect on 1 August 2026 [7]. Subordinate certificates signed from that date must carry an extended key usage extension and must not assert the catch-all value [7].

Current State

All four programs are moving against multi-purpose roots. Mozilla requires new roots to serve either websites or email, and existing dual-purpose roots must be split by 31 December 2028 [8]. Chrome requires hierarchies in its store to serve website authentication alone [4].

The policy texts themselves have been quiet since September. The Mozilla, Chrome, Apple and Microsoft policy repositories had no policy commits between 2 October and 25 October [9] [10]. Chrome Root Program Policy is still version 1.8, and Apple's newest commit is still from 16 October, with the policy at version 2.0 [55] [57]. The Mozilla, Microsoft and CCADB repository paths could not be read in the latest check. For those three, unchanged rests on the absence of any announcement on the public lists [36].

Chrome also publishes a statement of direction called Moving Forward, Together [12]. It names automation, term limits for roots, linting, stronger domain validation and shorter lifetimes as goals [12]. It carries no dates of its own, and its items bind only once they reach the policy or a Forum ballot [12].

Unresolved Conflicts

Two version records do not line up. One account attributes Chrome's June 2026 hierarchy rule to version 1.6, while the policy itself is version 1.8 dated 5 February 2026 [4]. Microsoft's requirements file carries a version 1.2 dated 20 May 2026, while its title and earlier records still describe version 1.1 effective 1 July 2026 [6]. Each program's own change history, read in order, would settle which rule entered at which version.

A third conflict concerns Mozilla's mass revocation rule. One source says plans must be tested each year by revoking 30 random certificates within five days, with attestation from April 2026

[13]. Mozilla’s template and its earlier guidance date third-party assessment from the June 2025 audit cycle [3] [14]. The current text of the policy’s revocation section would settle it.

2. The Post-Quantum Trust Path

A Merkle Tree Certificate replaces the usual chain of signatures with proof that the certificate appears in a signed public log, which keeps large post-quantum signatures out of most connections.

Recent Developments

Chrome and Apple are each designing a separate trust path for post-quantum certificates. Chrome’s draft quantum-resistant root program policy reached version 0.4.0 on 28 October and is designated the release candidate, labelled RC 1.0.0 [56]. It aligns with revision 07 of the IETF Merkle Tree Certificate draft [56]. The draft describes a root store made up entirely of Merkle Tree Certificate authorities and mirroring operators [15].

The new draft sets concrete limits. Each authority holds three to six ML-DSA-44 cosigner keys, one online and two offline in reserve [56]. Subscriber certificates last at most 7 or 47 days, depending on the issuing key, and reuse of domain validation is limited to 10 days [56]. These authorities may not issue subordinate certificates, and they are exempt from the annual audits under section 8 of the Baseline Requirements, though Chrome may ask for operational records instead [56].

One clause in the earlier 0.3.0 draft decided who could issue first. It limited initial eligibility to organisations that ran a usable Certificate Transparency log before 1 February 2026 [15]. The 0.4.0 text was read here only for the limits above, so whether that clause survived is not confirmed. Chrome opened applications to its testing programme on 31 July, with production planned for 2027 [11]. On 16 October it ruled that new cosigner requests must follow revision 07 of the IETF draft [11].

Apple announced its own Merkle Tree Certificate policy on 21 September, with a draft due by the end of October [16]. Each certificate would carry three cosignatures, one with ML-DSA and one with ECDSA from the issuer, and a third with ML-DSA from a separate mirror [16]. Validity is capped at seven days, and applications to run such an authority open in late summer 2027 [16]. Apple will also accept composite roots that pair ML-DSA with ECDSA or RSA for email [16].

Microsoft took a different route. On 27 August it deployed seven ML-DSA-87 pilot roots for approved authorities, including DigiCert, Sectigo and SSL.com [17]. These roots are for controlled testing only and are not publicly trusted [17]. They will not be added to the CCADB or to transparency logs [18].

A new issuer then applied. On 29 September Cloudflare said it intends to become a public certificate authority, buying root key material from GlobalSign in a deal it expects to close within two months [19]. It has applied to the Chrome, Apple, Microsoft and Mozilla programs [19]. It aims for inclusion in Chrome’s quantum-resistant store by early 2027, a store that does not yet exist in production [19] [11].

Current State

None of these plans is a published rule. Chrome's draft is a release candidate, not a policy [56]. Apple's date has slipped. On 30 October Apple wrote that "a WebPKI incident shifted the team's focus", so the draft will not appear by the end of October [16]. It will come "as soon as possible", with no new date [16].

The debate has begun anyway. On 29 September a participant on the transparency policy list argued that a seven-day lifetime cap is too large a change to bring in alongside Merkle certificates [16]. In that view it should be announced separately [16]. Chrome's 0.4.0 draft now allows 47 days for some keys, but the record does not say whether that answers the objection [56].

Chrome has already tied the programme to its security case. After the registries for three country-code domains were hijacked in early October, Chrome named shorter lifetimes, shorter validation reuse and the quantum-resistant programme as its answers [20]. It gave no new dates [20]. The certificate rules these designs depend on are covered in Parts 1 and 3.

3. Admissions and Removals

Inclusion requests and trust bit removals show who is entering the stores and who is leaving, and each one is argued in public before it happens.

Recent Developments

The first removal of the autumn was an emergency one.

On 20 October Mozilla removed emSign ECC Root CA - G3, an eMudhra root, from its root store after the private keys of eight intermediate authorities beneath it were found recoverable [21]. Chrome announced the same removal for 27 October, and its root store file now lists only one emSign root, which shows the removal shipped [22] [58]. Microsoft removed six eMudhra roots on 26 October [59]. Section 5 tells the story. eMudhra said on 25 October that it has decided to leave the public website certificate business [23].

The removal lands on an authority that was about to be admitted. eMudhra's four email and web roots had reached Mozilla's intent to approve on 2 October, with last call to 9 October [24]. One participant had questioned eMudhra's recent incident performance [24]. A separate bug had found seven undisclosed subordinate authorities run by a United Arab Emirates provider under one of eMudhra's roots [25]. The inclusion request still shows as pending approval, and no program has posted a decision on it [26] [76].

GoDaddy's two new web roots cleared public discussion on 29 September with no objections [27]. Mozilla posted its intent to approve on 2 October, then deferred approval until seven days after GoDaddy filed an updated practice statement [28]. GoDaddy published version 6.0 on 8 October, and on 15 October confirmed that the certificate profiles in that document did not match what it issues [29]. That admission led to the revocation in Section 6, which is now complete [29].

New applicants keep arriving. An inclusion request for two new roots from ISRG, the organisation behind Let's Encrypt, opened on 22 September [30]. IdenTrust opened the public

discussion of nine single-purpose roots on 9 October, to run until 20 November [31]. On 23 October IdenTrust reported, during that discussion, two new subordinate authorities whose profiles stray from its own policy, though not from the Baseline Requirements [32].

Two authorities asked to leave on their own terms. Certigna, which stopped issuing public website certificates in June, asked on 13 October for its websites trust bit to be removed [33]. Amazon Trust Services asked the same day for the Extended Validation trust bit to be removed from five of its roots [34].

Current State

No authority was admitted to any root store between September and 1 November, and one authority's root was removed. GoDaddy filed its full incident report on 29 October, but its inclusion request has not changed since 19 October and has no decision [66] [76]. The IdenTrust discussion has drawn no posts on the shared list since 9 October, and it is open until 20 November [36]. IdenTrust's bug on its two off-profile subordinate authorities has no comment after 23 October, so no revocation plan has been posted [75]. ISRG's two-root request and Amazon's Extended Validation request are unchanged [76].

The remaining eMudhra root is the one to watch. Chrome said it is "closely investigating the ongoing inclusion of eMudhra's remaining CA certificates" and promised updates on the CCADB public list, where none had appeared by 1 November [22] [36]. Mozilla removed only the one root, and emSign Root CA - G1 stays in its store [21] [60]. No program has decided on G1 [60].

One older question is open. In September a participant asked Mozilla what action it considers appropriate over NETLOCK's incident handling [37]. Mozilla replied on 5 October that it is weighing the concerns in its assessment of NETLOCK's continued inclusion, and will follow up publicly [37]. No later action was found.

The precedents are older. In April 2026 Mozilla and Chrome removed DigiCert's three G1 roots for age, not misconduct [38]. The punitive precedent remains the 2024 distrust of Entrust, which Chrome applied by issuance date rather than by revoking existing certificates [39]. The eMudhra removal is of a third kind, a protective removal after a compromise. Mozilla acted three days after the report, and Chrome and Microsoft within ten.

4. The Incident Record

Every failure to follow the rules is recorded as a public bug, and the quality of an authority's report is itself part of what the root programs judge.

Recent Developments

Twenty-one compliance bugs were opened between 24 October and 1 November, a quieter week than the 33 before it [61] [40]. Thirteen are unconfirmed and eight are assigned [61]. One theme runs through them, which is authorities blocking tools from fetching their revocation data by user agent. The first case was eMudhra's, and the pattern has since spread to others.

eMudhra's bug 2085612 got its full report on 29 October [52] [65]. A generic bot-protection profile, deployed on 28 September 2022, had blocked four common clients, and eMudhra

removed the filter on 26 October [65]. Izenpe's firewall returned a refusal to curl and Wget on its status responder, and Izenpe fixed it on 26 October [62]. Microsoft PKI Services reported that the Azure TLS issuing CA revocation list on www.microsoft.com returned a refusal to Wget [62]. Thirty authorities publish lists there, and Microsoft said a rule in place for over two years caused it, with an exception targeted for 3 November [62] [63].

Sectigo reported the same behaviour on 180 list addresses on its China hosts, fixed on 27 October [62]. Sectigo disputes that this breaches the Baseline Requirements wording on revocation lists [62]. Whether a refusal to common tools breaches the rules is therefore contested, and the user-agent bugs have not yet drawn a root program ruling.

Security and key handling produced three reports. TWCA self-reported on 29 October that indicators of unauthorized access were found on a Registration Authority server [64]. It isolated the server and said no mis-issuance has been identified to date [64]. TWCA also reported certificate serial numbers with fewer than 64 bits of random output, and Certigna reported a cross-signed root whose serial number came from an unseeded random generator [62].

Revocation list profile errors continue. HARICA published two truncated lists around 29 October and fixed them the next morning, with its full report due 11 November [62]. KIR's root list has a next-update time of exactly 12 months, which Let's Encrypt disputes as one second over the limit [62]. certSIGN also reported a list validity above the limit [62].

The week before had been heavier. Thirty-three bugs opened between 18 and 25 October, and the largest by count was TrustAsia's, with 19,237 certificates issued without honouring the domain owners' authorization records [40]. SHECA reported status responder keys with a public exponent of three, which HARICA called practically unrevocable [41]. SHECA revoked every authority under four of its roots that browsers do not trust and plans to destroy the keys [41].

DigiCert reported a revocation list whose sequence number went backwards and a list published three minutes late [42]. It also reported unauthorised access to a business system, saying "No certificate operations were impacted" [40]. Two authorities, Telekom's DFN and SwissSign, reported lists whose validity ran one second long, a defect the common linters accept [40]. Seven bugs in one week concern status responder certificate profiles, which looks like one third party's sweep [40].

September and early October had already been heavy. From 15 September the CCADB Policy required every authority's policy documents to state adherence to it, and the SHA-1 sunset required the revocation of SHA-1 subordinate certificates the same day [43] [44]. Four authorities missed that SHA-1 deadline [45] [46]. Thirty-eight bugs opened between 5 and 15 October, thirteen of them about failing status services, and ten more on 15 and 16 October [47] [48].

Current State

Follow-through on earlier bugs is uneven. TrustAsia's full report says a code change on 29 July let a record check for a parent domain be stored under a subdomain [69]. Of the 19,237 certificates, 194 were for domains whose records prohibited TrustAsia, and all were revoked by 18 October [69]. ACCV disclosed its replacement subordinate authority on 27 October and

revoked the old dual-purpose certificate on 29 October [68]. Distribution of the new hierarchy slipped from 31 October to 5 November [68].

DigiCert’s full report on the Cisco Meraki list defect arrived on 30 October [70]. It does not mention the 20-hour publication schedule DigiCert had promised, which DigiCert says was posted to the wrong incident, so that commitment is unconfirmed [70]. D-Trust filed a full report on 30 October on a finding that critical vulnerabilities were not patched within 48 hours between 30 June and 23 October [72]. It traces the cause to a ticketing workflow that was not updated and a departed security coordinator, and one action is due on 2 November [72]. For its second finding, on backup recovery testing, D-Trust asked for more time and promised an update on 6 November [73].

The late-audit bugs have not moved since CCADB staff acted on 21 October. The MULTICERT bug is assigned to Mozilla, and the three companion bugs against ITI Brazil, Consorci AOC and Entrust are untouched [71]. No new audit-related bug was filed in the latest week [71].

Mozilla’s public policy list carried the eMudhra compromise in October, with later threads on DNSSEC and on a proposed field for the domain validation method [49] [60]. The incident volume is high, and for the first time this autumn one incident has moved three root programs to act against an authority.

5. The eMudhra Key Compromise

A certificate authority’s intermediate keys are the keys that sign the certificates websites present, and if anyone else holds them, every browser that trusts the chain can be deceived.

Recent Developments

On Saturday 17 October at 18:05 UTC a security researcher told Chrome that the private keys of eight eMudhra intermediate authorities could be recovered [22]. Six issue website certificates and two issue code signing certificates trusted by Microsoft [21]. The keys came from a ceremony on 30 April 2025, and their private scalars were smaller than 256, so small that anyone could compute them [21]. Chrome described the cause as “insufficient randomness (i.e., weak cryptographic entropy)” [22].

The browsers moved within hours. Chrome corroborated the report and blocked the keys through its revocation list at 19:45 UTC, protecting its users without an update or restart [22]. Mozilla received the report at 20:06 UTC, recovered the keys itself at 21:48 UTC, and had its own block fully distributed by 22:59 UTC [21] [50]. Google’s Android block followed by Play Services on 19 October [22].

eMudhra revoked the eight certificates on the morning of 18 October, citing key compromise [21]. It wrote that “human errors in multiple steps” were to blame, and that its revocation lists had themselves been re-signed with the flawed keys [21]. Sectigo revoked its eight cross-certificates for the same authorities on 19 October [21].

Then came the removals. On 20 October Mozilla wrote that “the circumstances warrant the emergency removal of trust in emSign ECC Root CA - G3”, the root that signed the eight [21]. Its reason was that the keys had by then been recovered publicly by several people [21]. NSS

3.131.1 removed that root the same day, along with two older eMudhra roots already scheduled to go, and Firefox 158.0.1 and two extended support releases carried it [51] [50].

Chrome scheduled its removal of the same root for Tuesday 27 October [22]. Its root store file is now at version 44 and lists only emSign Root CA - G1 among the emSign roots [58]. The file carries no date or changelog, so the 27 October date is an inference from the root's absence. About 30,000 unexpired certificates chained to the root, almost all issued through ACME, and eMudhra stopped issuing under it that afternoon [23].

Microsoft published its own response on 26 October, revised the next day [59]. It covers the same eight subordinate authorities, six for websites and two for code signing, and it calls the event a third-party incident, not a compromise of Microsoft systems [59]. On 20 October a Windows out-of-band update blocked the eight keys, and Edge received an emergency revocation list update [59]. On 26 October another Windows update removed six emSign G3 roots and Sectigo's AAA Certificate Services cross-signature from the trust list [59]. Edge removed the same seven certificates on 27 October [59].

Chrome said it had "reviewed and confirmed that no other keys included in our root stores share the same failure mode with weak entropy" [22]. On 27 October it added that a scan of all Chrome-trusted certificates found no further weak keys [60]. It also filed a change to the CCADB tools that would gate disclosure on weak-key checks [60]. No transparency log held a website certificate from the affected authorities before the block, so no forged certificate could have worked against clients that enforce logging [22].

Current State

eMudhra is leaving. At 03:53 UTC on 25 October its representative wrote that the company "has decided to exit public TLS", with a full report to follow within the 14-day window [23]. That window ends about 1 November, and the full report on the key compromise had not been posted by then [60]. A reporter also wrote that a preliminary report on the affected revocation list, due 29 October, was never filed, and that a second list was affected [60].

Thirty minutes after that statement a reply counted 107,107 certificates logged on 20 and 21 October under the revoked keys [23]. Of these, 106,050 were proofs of concept carrying a Google name [23]. eMudhra's side bugs have moved. It posted full reports on 8,288 certificates missing its certificate-type identifier, on four outdated practice statements, and on 41 revoked status responder certificates [74]. The refusal of common command-line tools by its status servers became the separate bug covered in Section 4 [65].

The questions about the compromise itself are outstanding. Chrome posted twelve on 21 October, and eMudhra attached a written response on 29 October that it says it also shared directly with the browser programs [60]. That response was not read here. Mozilla's written questions of 20 October were unanswered on 23 October [23]. Mozilla's only later action was to mark two bugs as duplicates [60].

The open decision is emSign Root CA - G1, which stays in Chrome's and Mozilla's stores. No program has decided on it [60]. eMudhra says its G1 roots "remain active and continue to be fully supported" and that it will stop issuing website certificates from G1 once a federation's

transition is complete [60]. One commenter wrote that there is no basis for relying parties to keep trusting G1 [60]. Apple has made no statement that could be found.

Java is an unverified lead. A search points to an Oracle security alert dated 21 October that distrusts the compromised intermediates, but all three Oracle hosts refused access, so it is not stated as fact [77]. Part 5 covers what the flood of proof-of-concept certificates did to the transparency logs, and Part 1 covers the draft rule on small private scalars that followed within five days.

Unresolved Conflicts

The count of affected authorities is given two ways. Both root programs say eight keys were compromised, six for websites and two for code signing [21] [22]. Chrome's note on transparency logging speaks of "these six intermediate CAs", which is the website subset [22]. The two numbers describe the same event at different scopes, and eMudhra's full report should list all eight.

6. GoDaddy's Mass Revocation

A mass revocation is the moment a root program's rules reach ordinary websites, because every affected certificate must be replaced by its owner.

Recent Developments

On 15 October GoDaddy opened an incident for 404,607 certificates [53]. A third party had found that the policy information in certificates from its GoDaddy and Starfield hierarchies did not match its certificate policy and practice statement [53]. The Starfield hierarchies hold almost all of them, with 208,896 under one domain validation profile and 187,159 under another [53]. GoDaddy stopped issuance, corrected the profiles and resumed issuance on the corrected ones [53].

The revocation deadline was 18 October at 23:45 UTC [54]. On 16 October GoDaddy said it did not expect to meet it, and filed the delay as a separate incident [54]. By the deadline it had revoked 170,566 certificates, and by 05:29 UTC on 19 October the total was 227,422 [54].

Current State

The work is done, four days late. On 22 October GoDaddy reported that revocation was complete as of 22:50 UTC that day [54]. Its corrected total was 403,643 certificates, of which 378,067 were revoked and 25,576 had expired on their own, leaving none valid [54].

The full incident report arrived on 29 October [66]. It dates the non-compliance from 28 February 2025 to 14 October 2026 and confirms 403,643 affected certificates, none valid after 22 October [66]. The cause was incorrect profile information introduced in two practice statement updates, versions 5.02 and 5.03, with no check before publication and no automated linting against the stated profiles [66]. Action items run from 4 December 2026 to 15 January 2027 [66].

No root program has commented on either GoDaddy bug, and the delayed revocation bug has no comment after 22 October [67]. CCADB staff confirmed the original mis-issuance bug on 21

October [53]. GoDaddy’s root inclusion application remains open and unchanged since 19 October [35] [76].

Unresolved Conflicts

How Mozilla weighs the delay against the pending inclusion is open. A decision posted on the inclusion request, or a root program comment on the delayed revocation bug, would settle it.

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